

Toward Understanding the Return Path Capacitance in Capacitive Human Body Communication

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Abstract—Human Body Communication (HBC) utilizes the human body as a conductive channel to enable ultra-low-power secure communication between devices on and around the body. In Capacitive HBC, although the body provides the forward path for communication, the closed loop return path is formed through parasitic coupling of the device's ground plane with the environment. This return path capacitance (C_G) is a critical factor in determining the overall HBC channel response. Hence an in-depth understanding C_G and the key factors affecting its value will enable optimized design of HBC systems. In this brief we provide a detailed analysis of the return path capacitance for wearable devices of different shapes (disc, rectangular) through derivations, simulations as well as experiments. FEM simulation results show that the parasitic capacitance is almost equal to the self capacitance found through theoretical derivations, when the distance of the device to earth's ground is significantly larger than the device dimensions. Independent experiments using an LCR meter and wearable devices, show close correspondence with theory and simulations. The return path capacitance (C_G) value, which is in the range of a few pFs for a watch sized wearable, is primarily dependent on the size of the device and almost independent of the distance from earth's ground.

Index Terms—Capacitive human body communication, human body channel modeling, return path capacitance, body area network, wearable electronics.

I. INTRODUCTION

HUMAN body communication (HBC) is an emerging technology with wide-ranging potential to transform neuroscience, connected healthcare, human-computer interaction among others. It proposes [1]–[6] to use the human body as a conductive medium to achieve connectivity around the body at $\sim 100\times$ lower energy [4] compared to radio wave based Wireless Body Area Network (WBAN), while simultaneously making it more physically secure.

There are two dominant modalities of HBC that can be found in literature - capacitive and galvanic. In capacitive HBC, as shown in Fig. 1(a), a single ended electrode at the transmitter end is used to couple the signal onto the human

body which then propagates through the human body and gets picked up at the receiver using another single ended electrode. This only creates the forward path of signal transmission. The return path to close the loop in this case is formed using parasitic capacitance C_G , between the on-board ground plate and earth's ground at both the transmitter and receiver ends [7]. Particularly in frequencies less than 1 MHz, HBC operates in the Electro-Quasistatic regime and there will be no communication without a return path. In galvanic HBC, the signal is applied onto the human body using two electrodes at the transmitter, which forms its own closed loop [8], [9]. The “fringe-fields” then travel throughout the body and is picked up differentially using two other electrodes at the receiver end. Hence, galvanic HBC does not depend on the parasitic capacitive return paths strongly. On the other hand, in capacitive HBC, the return path is the most important determining factor for channel loss [1] and is given by $(C_G/C_{Body}) \cdot (C_G/C_L)$ (Fig. 1(a) and (b), detailed discussion in Section II-A) where C_{Body} is the parasitic capacitance between the body and the surrounding environment and C_L is the termination capacitance at the receiver end. (Terminating with capacitance minimizes low frequency loss and provides a wide-band channel for communication [1], [4] as shown in Fig. 1(c). Increasing C_G or reducing C_L , reduces absolute channel loss.)

At low frequencies, the series impedance R_B provided by the human body (Fig. 1) is negligible compared to the impedance formed by the return path capacitance, C_G . Hence, in capacitive HBC, the HBC channel loss is only a weak function of the distance between the transmitter and the receiver on a human body, making it preferable over galvanic HBC for longer distance communication. Further, it can be noticed that C_G is the primary factor for determining the HBC channel loss for a fixed C_L (Fig. 1(c)). While there have been many experimental studies for capacitive HBC channel loss measurements [1], [5], [9]–[11] there has been no comprehensive analysis targeted at understanding the origin and variation of C_G . In this brief, we discuss for the first time a theoretical formulation of C_G validated by simulations and experiments, to allow for better informed design of HBC systems.

II. THEORETICAL FORMULATION

In this section, we calculate an expression for HBC channel loss to motivate the importance of return path capacitance C_G in HBC and then explore an equation based model for variance of C_G with respect to distance of an HBC device with earth ground.

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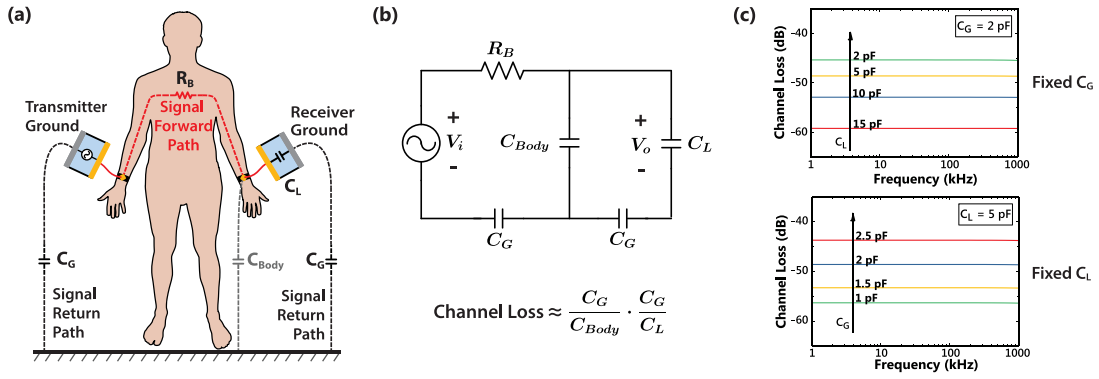


Fig. 1. (a) Illustration of capacitive HBC, where signal return path between transmitter and receiver is formed by parasitic capacitance between their respective ground plates and earth's ground. (b) Simplified circuit model for calculating HBC channel loss, with typically $R_B = 10k\Omega$, $C_{Body} = 150 \text{ pF}$ [1]. (c) Frequency response of HBC channel loss, showing dependencies of channel loss on C_G and C_L .

A. Derivation of Channel Loss From Simplified Channel Model

The simplified circuit model for the human body channel characteristics is shown in Fig. 1. The output voltage (V_o) and the input voltage (V_i) are related to each other as follows (neglecting the forward path body resistance R_B):

$$\frac{V_o}{V_i} = \frac{C_G}{C_G + C_{Body} + \frac{C_L C_G}{C_L + C_G}} \cdot \frac{C_G}{C_L + C_G} \quad (1)$$

where C_G is the return path capacitance, C_{Body} is the capacitance between the body and the ground, C_L is the load capacitance. The forward path resistance (R_B) is neglected in this calculation, as R_B is of the order of $10K\Omega$, whereas the impedance provided by the return path capacitance is greater than $100K\Omega$ for frequencies less than 1 MHz. Further, the return path capacitance (C_G) is an order of magnitude smaller than the load capacitance (C_L , order of 10 pF) which in turn is an order of magnitude smaller than the body to ground capacitance (C_{Body} , $>100 \text{ pF}$) [1]. Hence we can make the following approximation:

$$C_G + C_{Body} + \frac{C_L C_G}{C_L + C_G} \approx C_{Body} \quad (2)$$

Substituting equation (2) in equation (1), a simplified expression for the human body communication channel loss (V_o/V_i) can be obtained as

$$\frac{V_o}{V_i} \approx \frac{C_G}{C_{Body}} \cdot \frac{C_G}{C_L} \quad (3)$$

Equation (3) shows that the channel loss for communication between two wearable devices is dependent on C_G^2 . This shows the importance of determining the value of return path capacitance to estimate the HBC channel loss and to be able to design optimized HBC systems.

B. Variation of C_G With Distance From Earth Ground

The ground plate of an HBC device, e.g., a smartwatch, can be modelled as a disc to analyze the return path capacitance to the earth's ground. If ground is placed really close to the object, parallel plate approximation holds for capacitance

(Fig. 2(a)). On the contrary, if the ground is placed far away from the object, e.g., a concentric sphere and a spherical shell shown in Fig. 2(b), capacitance is distance dependant, and reduces to the *self-capacitance* of the sphere as $d \rightarrow \infty$. But what is the capacitance of any object placed at a finite distance from an infinite ground plane? To investigate this, we start with a spherical conductor for simplicity and obtain an analytical expression using image charge analysis [12]. Let us assume a sphere of radius a with a surface potential V_{sphere} (Fig. 2(c)) with it's center at a distance d from an infinite ground plane, i.e., a plane at zero potential.

- 1) To replace the system by a set of image charges, a charge q_1 is placed at the center of the sphere such that $V_{sphere} = q_1/4\pi\epsilon_0 a$. This creates an equipotential at the surface of the sphere.
- 2) Next, a charge $q'_1 = -q_1$ is placed at a distance d below the ground plane. This creates a zero potential plane at the ground surface, but destroys the equipotential at the spherical surface.
- 3) To recover the spherical equipotential, an image charge of q'_1 , $q_2 = (aq_1)/2d$ is placed at a distance $a^2/2d$ below the center of the sphere. This recreates the spherical equipotential by canceling the effect of q'_1 , in turn destroying the zero potential plane at the ground.
- 4) This process is iterated, resulting into an infinite series of image charges, with the magnitude of the image charges diminishing in each subsequent iteration (Assuming $a < d$, which is trivially true.)

The resulting charge distribution indicates a total charge of $Q = \sum q_i$ charge on the sphere. Writing the charges as a function of q_1 ,

$$Q = q_1 + \frac{aq_1}{2d} + \frac{2adq_1}{4d^2 - a^2} + \dots = C_{sphere} \frac{q_1}{4\pi\epsilon_0 a} \quad (4)$$

using $Q = C_{sphere} V_{sphere}$. So, C_{sphere} can be calculated as:

$$C_{sphere} = 4\pi\epsilon_0 a \left(1 + \frac{a}{2d} + \frac{2ad}{4d^2 - a^2} + \dots \right) \quad (5)$$

When $d \gg a$, this expression reduces to

$$C_{sphere} = 4\pi\epsilon_0 a \quad (6)$$

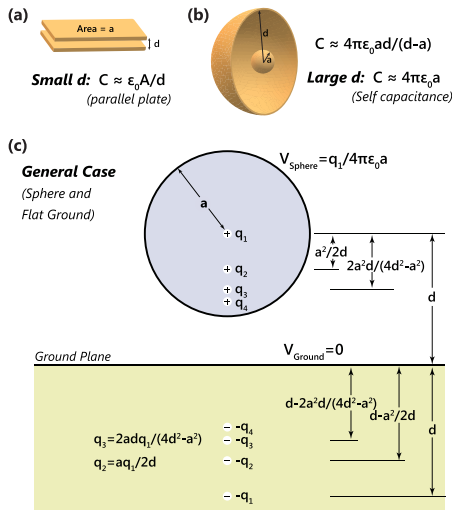


Fig. 2. Capacitance between two objects for (a) small distance, and (b) large distance. (c) Image charge setup for general case: a conducting sphere near a ground plane. An infinite sequence of image charges are required to replicate the equipotential at the surface of the sphere along with the zero potential ground plane.

which is self-capacitance of the sphere. For a disc of thickness h and radius a , it's self capacitance can be shown to be [13]:

$$C_{disc} = 8\epsilon_0 a \left[1 + 0.87 \left(\frac{h}{2a} \right)^{0.76} \right] \quad (7)$$

so that for a thin disc, when $h \ll a$,

$$C_{disc} \approx 8\epsilon_0 a \quad (8)$$

Now the exact closed form solution for the decay of capacitance in the case of a disc may be different, but the trend of decay should be similar to that of a sphere. To demonstrate this, electrostatic FEM based simulations are performed in ANSYS Maxwell to plot the capacitance of a sphere of radius 10 cm and a disc of the same radius and thickness 1 cm with a finite ground plate (to restrict simulation space), as shown in Fig. 3. For both the sphere and disc, the simulated capacitances decay to their self capacitances given by equations (6) and (7) respectively. *The percentage offset from the self-capacitance is less than 5% for $d/a > 10$.* So for design purposes, the self-capacitance of the ground plate of a device should provide a good estimate for the return path capacitance. As long as the device dimension is small compared to its distance to earth's ground, it's orientation or distance with respect to ground will not matter significantly.

III. EXPERIMENTS AND DISCUSSION

A. Verifying Trend of C_G With Distance by Using an LCR Meter

To validate the formulation of capacitance decay with distance (Section II-B), we experimentally find out the capacitance of a disc shaped conductor with a ground plate at various distances. The experimental setup is shown in Fig. 4(a). A disc shaped conducting plate is fashioned out of a conducting copper tape, and a steel plate is used as a ground plate. The capacitance between these two objects is

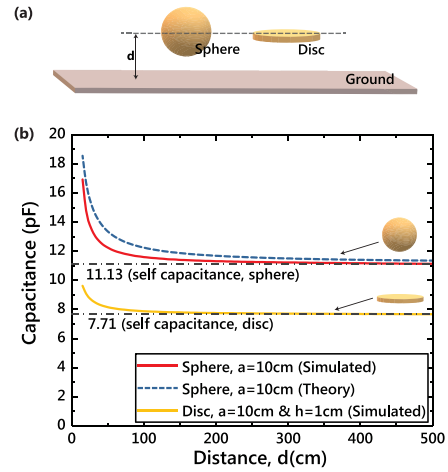


Fig. 3. (a) Simulation schematic and (b) plot of capacitance of a sphere and a disc vs distance to ground plane, obtained through simulations. For the sphere, the simulated result is compared with the response given by equation (5). Note that the simulated response drops faster to the self capacitance, as a finite ground plate is used for the simulations.

measured using an LCR meter (used to measure Inductance (L), Capacitance (C) and Resistance (R)), at different distances between the disc and the plate. For each measurement the LCR meter is calibrated to ensure that the capacitances between the measurement leads and the wires connecting the two plates do not affect the results. The measured capacitance trend, as shown in Fig. 4(b), closely follows the simulated variation of capacitance with distance.

It is to be noted that in our experiments, the ground plate connected to the LCR meter is finite sized unlike the earth's ground. The capacitance between two finite sized objects should reduce to zero as the distance between them goes to infinity. However, both the disc and the ground plate in our experiment also couple to the earth's ground - the floor and the walls of the room in this case - through parasitic capacitance. These spurious capacitances become dominant in determining the net coupling capacitance between the disc and the plate when they are spaced far apart and render the capacitance between those objects independent of distance beyond a certain threshold. So, the capacitance between these two objects does become constant at large distance in our experiment, exhibiting the same trend discussed in the previous section. This can be seen in the results from the LCR meter measurements in Fig. 4(b), where the measured capacitance saturates beyond a separation of around 60 cm. We repeat the experiment by replacing the copper disc by a micro-controller PCB, where the PCB is supposed to emulate a real-life HBC device. There, we measure the capacitance between the ground plane of the PCB, and the steel plate. The resulting trend, as seen in Fig. 4(c), is similar to that of the disc.

B. Measurement of C_G for Wearable Devices

We carry out experiments with a wearable transmitter and receiver to estimate C_G in a wearable-to-wearable communication scenario. The schematic of the experimental setup is

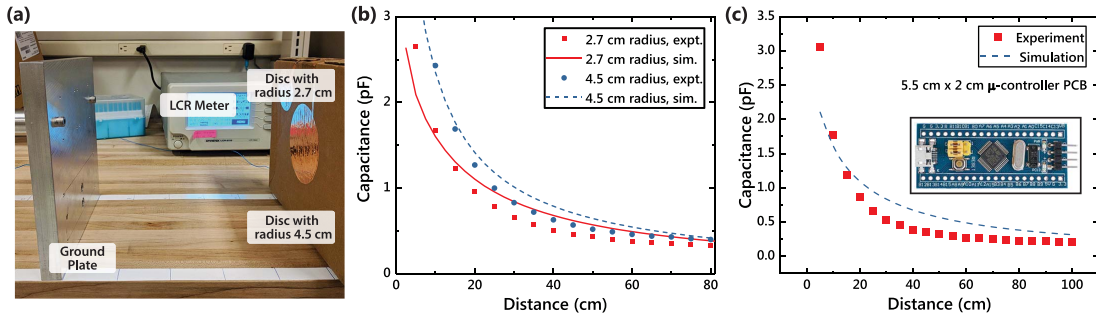


Fig. 4. (a) Experimental setup for capacitance measurements between a disc formed out of copper tape and a steel ground plate using an LCR meter (b) Plot of measured capacitance, as the distance between the ground plate and the disc is varied. The capacitance value shows dependence on the size of the disc. (c) Same experiment as (b), except the disc is replaced by a $5.5 \text{ cm} \times 2 \text{ cm}$ micro-controller PCB (shown in inset). The measured capacitance is that between the steel ground plate, and the ground plane of the PCB.

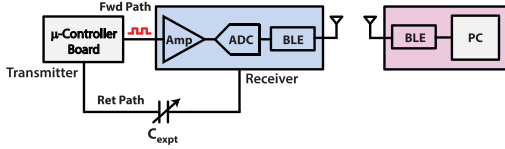


Fig. 5. Schematic of the experimental setup used to calibrate and measure return path capacitance of a micro-controller board in a wearable-to-wearable scenario. Any effect introduced from LCR meter in the outcome of the previous measurements is absent in this setup, as the output voltage is recorded at the receiver itself and transmitted wirelessly to a computer. The circuit theoretic model for this setup is shown in Fig. 6(a).

shown in Fig. 5. To get rid of human body channel characteristics and eliminate C_{Body} , the transmitter and receiver are electrically connected by a conducting wire in this setup whereas in an actual HBC communication, the forward path between the transmitter and the receiver would be formed through the human body. Further, the ground pins of the receiver and the transmitter devices are connected through a known capacitance C_{expt} . The return path of the signal then constitutes of C_{expt} , in parallel with C_G and an additional capacitance C_{setup} induced by the wires in the return path. This results into the circuit model shown in Fig. 6(a) for the measurement setup, and the relation between the output and the input voltage can be obtained as

$$V_o = \frac{C_G + C_{\text{setup}} + C_{\text{expt}}}{C_G + C_{\text{setup}} + C_{\text{expt}} + C_L} A_v V_i \quad (9)$$

where V_o : voltage at the receiver, A_v : voltage gain of the amplifier, V_i : input signal, C_G : return Path capacitance, C_{expt} : external capacitance introduced for the return path measurements, C_{setup} : extra capacitance introduced in the measurement from the experimental setup, C_L : load capacitance at the receiver end.

To measure the amplitude of the received signal accurately, standard bench-top instruments such as an Oscilloscope or a Vector Network Analyzer (VNA) cannot be used, as these instruments introduce extra capacitance in the return path due to their large internal ground plates and end up affecting the measurement results. To realize a truly miniaturized wearable measurement setup, the signal amplitude needs to be measured at the receiver node itself. This is achieved by using an on-board Analog to Digital Converter (ADC) of a

STM32F103C8 microcontroller built into the receiver. The signal at the receiver end is amplified using an amplifier of gain A_v before being applied to the ADC, to ensure that the weak input signal can be measured accurately over the inherent noise of the ADC. The input signal provided is a square wave and the amplitude at the receiver end is calculated by taking the difference between consecutive samples and plotting a histogram of the differences. To enable this processing, data obtained from the ADC is sent via Bluetooth to a computer. This way we can *unobtrusively* measure the signal corresponding to an actual HBC scenario and get a proper estimation of the return path capacitance. The closed loop gain A_v of the amplifier was characterized to be ~ 9 and the resolution of the ADC was 0.8 mV. V_o is measured for multiple values of C_{expt} , as well as by shorting the return path to determine $A_v V_i$ (Fig. 6(b)). By rewriting equation (9) as

$$\frac{V_o}{A_v V_i - V_o} = \frac{C_G + C_{\text{setup}}}{C_L} + \frac{C_{\text{expt}}}{C_L} \quad (10)$$

it can be seen that a graphical plot of the ratio $V_o/(A_v V_i - V_o)$ vs the capacitance C_{expt} will have a slope $1/C_L$ (Fig. 6(c)). Note that this slope $1/C_L$ is independent of the amplifier gain A_v , as the gain is cancelled out when we take the ratio of V_o and $(A_v V_i - V_o)$ in equation (10). So using equation (10), value of C_L can be determined. Once C_L is known, open ground measurements are carried out by disconnecting the ground pins of the transmitter and the receiver, so that the return path constitutes of only C_G . In this setup, $C_{\text{setup}} = C_{\text{expt}} = 0$, and equation (10) reduces to:

$$\frac{V_o}{A_v V_i - V_o} = \frac{C_G}{C_L} \quad (11)$$

So using the previously determined value of C_L , C_G can be estimated. Using this method, the return path capacitance of the microcontroller board in use was calculated to be 1.9 pF, with the transmitter-receiver setup placed 1 meter away from the floor. The ground plane of the receiver has dimensions of $3.6 \text{ cm} \times 8.5 \text{ cm}$, which translates to the same surface area as a disc of radius 3.1 cm. Using $a = 3.1 \text{ cm}$ in equation (8), self capacitance of the disc turns out to be 2.2 pF, which is close to the experimentally obtained C_G of 1.9 pF for the microcontroller board. So equation (8) (self capacitance of a thin disc) can indeed be used to get a reasonable estimate of

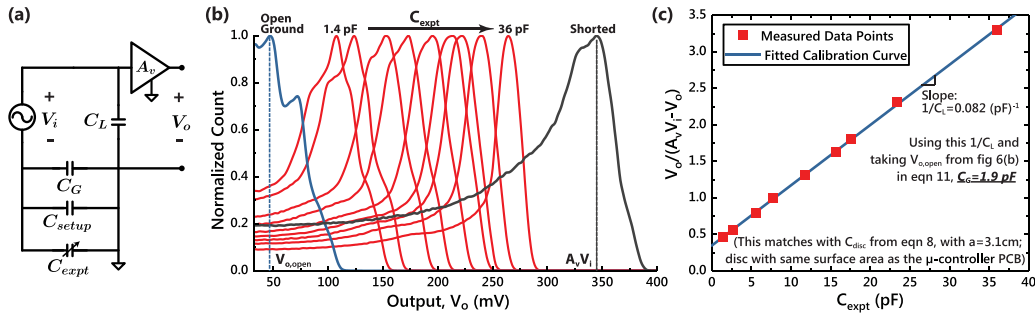


Fig. 6. (a) Circuit model for wearable-to-wearable return path capacitance measurement setup. (b) Measured signal amplitude histogram of amplifier for different values of C_{expt} , as well as open and shorted return paths. For open return path, $C_{setup} = C_{expt} = 0$ and for shorted return path, $V_o = A_p V_i$. (c) Calibration curve obtained, for prediction of C_G through linear fit of the histogram peaks in (b) against C_{expt} . Using the obtained C_L and the open ground histogram peak, the value of C_G is determined.

C_G of an HBC device. As for most typical wearable devices, the value of a lies in the range of 1.5 cm to 3 cm (size of a watch), the typical C_G would lie in the 1 – 2 pF range. Using the circuit from Fig. 1(b), typical values [1] of $R_B = 10 \text{ k}\Omega$, $C_{Body} = 150 \text{ pF}$ and $C_L = 5 \text{ pF}$ and varying C_G between 1 – 2 pF, circuit simulation shows 45 – 55 dB channel loss (Fig. 1(c)). This supports experimental works in capacitive HBC [1], [14], and hence validates the accuracy of our model.

It is to be noted that these experiments have been performed in the absence of a human subject, to eliminate the effect of human body as a forward path. In actuality, if an HBC device is close to the torso and the body is functioning as the forward path, then some electric field lines from the ground plane of the HBC device would shunt to the body, reducing the “effective C_G ”. If the torso is not close, presence of wrist would cause only about 10% reduction from ideal C_G . On the contrary, if another human subject comes close to the person using an HBC device, a lower impedance path will be formed between the HBC device’s ground and earth’s ground - increasing the “effective C_G ” from self-capacitance. Despite these deviations, the self-capacitance provides a good first-order approximation of C_G , and can be used effectively for design purposes.

IV. CONCLUSION

In summary, we provide a detailed analysis of return path capacitance, C_G in capacitive HBC. We postulate that C_G is close to the self capacitance of the ground plate of an HBC device, provided the dimensions of that device is small compared to its distance from earth’s ground. This is then validated through simulations and two different sets of experiments. We propose an experiment to measure C_G of an wearable HBC device. We show that C_G turns out to be in the 1 – 2 pF range for a wearable of the size of a watch and that C_G can be estimated from the self capacitance of a disc of similar dimension, reiterating that C_G is dependent on the size of the device and almost independent of its distance to earth’s ground. Concisely, estimating C_G as self capacitance for $d \gg a$ through equations (7) or (8) allows us to estimate the HBC channel loss $((C_G/C_{Body}) \cdot (C_G/C_L))$ and enables physics-based, model-driven design and optimization

of Human Body Communication systems, instead of empirical designs that has been the norm.

REFERENCES

- [1] S. Maity, M. He, M. Nath, D. Das, B. Chatterjee, and S. Sen, “Bio-physical modeling, characterization, and optimization of electro-quasistatic human body communication,” *IEEE Trans. Biomed. Eng.*, vol. 66, no. 6, pp. 1791–1802, Jun. 2019.
- [2] D. Das, S. Maity, B. Chatterjee, and S. Sen, “Enabling covert body area network using electro-quasistatic human body communication,” *Sci. Rep.*, vol. 9, no. 1, p. 4160, 2019. [Online]. Available: <https://doi.org/10.1038/s41598-018-38303-x>
- [3] T. G. Zimmerman, “Personal area networks: Near-field intrabody communication,” *IBM Syst. J.*, vol. 35, no. 3.4, pp. 609–617, 1996.
- [4] S. Maity, B. Chatterjee, G. Chang, and S. Sen, “A 6.3pj/b 30Mbps –30dB SIR-tolerant broadband interference-robust human body communication transceiver using time domain signal-interference separation,” in *Proc. IEEE Custom Integr. Circuits Conf. (CICC)*, San Diego, CA, USA, Apr. 2018, pp. 1–4.
- [5] J. Park, H. Garudadri, and P. P. Mercier, “Channel modeling of miniaturized battery-powered capacitive human body communication systems,” *IEEE Trans. Biomed. Eng.*, vol. 64, no. 2, pp. 452–462, Feb. 2017.
- [6] H. Cho *et al.*, “A 79 pJ/b 80 Mb/s full-duplex transceiver and a 42.5 μW 100kb/s super-regenerative transceiver for body channel communication,” in *IEEE Int. Solid-State Circuits Conf. (ISSCC) Dig. Tech. Papers*, Feb. 2015, pp. 1–3.
- [7] S. Maity, K. Mojabe, and S. Sen, “Characterization of human body forward path loss and variability effects in voltage-mode HBC,” *IEEE Microw. Compon. Lett.*, vol. 28, no. 3, pp. 266–268, Mar. 2018.
- [8] M. S. Wegmueller, M. Oberle, N. Felber, N. Kuster, and W. Fichtner, “Signal transmission by galvanic coupling through the human body,” *IEEE Trans. Instrum. Meas.*, vol. 59, no. 4, pp. 963–969, Apr. 2010.
- [9] M. A. Callejón, D. Naranjo-Hernández, J. Reina-Tosina, and L. M. Roa, “A comprehensive study into intrabody communication measurements,” *IEEE Trans. Instrum. Meas.*, vol. 62, no. 9, pp. 2446–2455, Sep. 2013.
- [10] Ž. Lucev, I. Krois, and M. Cifrek, “A capacitive intrabody communication channel from 100 kHz to 100 MHz,” in *Proc. IEEE Int. Instrum. Meas. Technol. Conf.*, May 2011, pp. 1–4.
- [11] J. A. Ruiz and S. Shimamoto, “A study on the transmission characteristics of the human body towards broadband intra-body communications,” in *Proc. 9th Int. Symp. Consum. Electron. (ISCE)*, Jun. 2005, pp. 99–104.
- [12] D. J. Griffiths, *Introduction to Electrodynamics*. London, U.K.: Pearson Education Inc., 2011, ch. 3.2, pp. 121–126.
- [13] W. R. Smythe, “Charged right circular cylinder,” *J. Appl. Phys.*, vol. 33, no. 10, pp. 2966–2967, 1962. [Online]. Available: <https://doi.org/10.1063/1.1728544>
- [14] S. Maity, D. Das, and S. Sen, “Wearable health monitoring using capacitive voltage-mode human body communication,” in *Proc. 39th Annu. Int. Conf. IEEE Eng. Med. Biol. Soc. (EMBC)*, Jul. 2017, pp. 1–4.